Release Notes for USB2ANY.DLL

v2.6.3.0

Bug fixes, new features, etc.:

1. Debug settings for Debug Logging, Packet Logging, and Profiling are saved/restored to/from a settings file specific to the application program that loaded USB2ANY.DLL. This allows the settings to be remembered from session to session.
2. The current default Debug Logging, Packet Logging, and Profiling settings are now logged to the debug log at the beginning of each session.
3. Added code to display a message box, with a performance warning, at the beginning of a session if Packet Logging and/or Profiling are enabled.
4. The full path of the program that loaded USB2ANY.DLL is now logged to the debug log at the beginning of each session.
5. Added new API function, u2aUART_SetMode(), to allow the setting of special UART modes.
6. Added UART_Mode enums UART_Normal, UART_ReceiverOff, and UART_RecvAfterXmit.
7. Some improvements to Packet Logging.
8. Some improvements to Debug Logging of UART functions.

v2.6.2.25 (beta)

Bug fixes, new features, etc.:

9. Added source file USB2ANY_Firmware.cpp (generated by ConvertUSB2ANYHexFile.exe).
10. Many improvements and bug fixes to the firmware loader.
11. USB2ANY.DLL now requires a matching version of firmware.
12. The required firmware version is now embedded into USB2ANY.DLL.
13. Loader dialog now displays "Please wait..." while waiting for BSL mode.
14. USB2ANY.DLL no longer depends on BSL.DLL (the required code is now built-in).

Other changes:

15. Added GetRequiredFirmwarePath().
16. In ParseFirmwareVersion(), dwVersion is updated only if it is not NULL.
17. Minor improvements to RunProcess().
18. Added CalcCRC().

v2.6.2.20 (beta)

Bug fixes, new features, etc.:

1. Added source file USB2ANY_Firmware.cpp (generated by ConvertUSB2ANYHexFile.exe)
2. Added source files BSLButtonHelp.h and BSLButtonHelp.cpp to display BSL help.

3. Improved code used to locate the SDK folder. It now works correctly when the SDK is installed into a non-standard location.
4. Added CU2AClass::EasyScale_WriteAndRead() function.
5. Added CU2AClass::OneWire_PulseWriteEx() function.
6. Marked OneWire_Channel enums as deprecated.
7. Deprecated OneWire_Mode enum OW_EnablePulseMode.
8. Added OneWire_Mode enums OW_Mode1, OW_Mode2, and OW_Mode3.
9. Added OneWire_Flags enums OW_FLAG_NONE, OW_FLAG_INVERT, and OW_FLAG_NO_INIT.
10. Added u2aSMBUS_BlockWriteBlockRead() function.
11. Added u2aI2C_BlockWriteBlockRead(0 function.
12. Added u2aEasyScale_WriteAndRead() function.
13. Added u2aOneWire_PulseWriteEx() function.
14. Added u2aUART_GetRxCount() function.
15. Increased range of *milliseconds* parameter for SetReceiveTimeout() function from 2 - 100 to 2 - 200 .
16. Added U2A_ApiClassV2::I2C_BlockWriteBlockRead() function.
17. Added U2A_ApiClassV2::UART_GetRxCount() function.
18. Added U2A_ApiClassV2::EasyScale_WriteAndRead() function.
19. Added U2A_ApiClassV2::OneWire_PulseWrite() function.

Other changes:

20. Added code to prepare for static firmware loader.
21. Added Cmd_EasyScale_WriteAndRead.
22. Modified to use version string from Version.h (several places).
23. Added source files ReceiveTimeoutDialog.h and ReceiveTimeoutDialog.cpp to display a dialog that allows the user to change the receive timeout.

v2.6.2.9 (beta)

Bug fixes, new features, etc.:

1. Added API function u2aOneWire_PulseSetup
2. Added API function u2aOneWire_PulseWrite
3. Added API function u2aOneWire_SetMode
4. Added API function u2aOneWire_SetState
5. Added ERR_NOT_ENABLED error code
6. Added example code for OneWire interface
7. Added OneWire tab to USB2ANY Explorer
8. Fixed bug that caused the wrong baud rate to be logged
9. Fixed some baud rate selection issues
10. Added "BSL Help" button to Firmware Update dialog

Other changes:

11. Added CU2AClass::OneWire_PulseSetup()
12. Added CU2AClass::OneWire_PulseWrite()
13. Added CU2AClass::OneWire_SetMode()

14. Added CU2AClass::OneWire_SetOutput()
15. Added Cmd_I2C_BlkWriteBlkRead
16. Added Cmd_OneWire_PulseSetup
17. Added Cmd_OneWire_PulseWrite
18. Added Cmd_OneWire_SetMode
19. Added Cmd_OneWire_SetState
20. Added OneWire_Mode and OneWire_Channel typedef enums
21. Changed Cmd_I2C_Read_Internal to Cmd_I2C_ReadInternal
22. Changed Cmd_I2C_Write_Internal to Cmd_I2C_WriteInternal
23. Added new version string macros
24. Added code to UART_Read to attempt read multiple times
25. Added U2A_ApiClassV2::OneWire_PulseSetup()
26. Added U2A_ApiClassV2::OneWire_SetMode()
27. Added U2A_ApiClassV2::OneWire_SetOutput()
28. Changed Cmd_DigitalCapture to Cmd_Digital_Capture
29. Added PinManager code for OneWire
30. Added support for OneWire interface
31. Added u2aOneWire_PulseSetup()
32. Added u2aOneWire_SetMode()
33. Added u2aOneWire_SetOutput()
34. Added code to display the BSLButtonLocator dialog when the help button is clicked

v2.6.2.0

1. Added u2aSuppressFirmwareCheck() function
2. Changed splash screen code to not display splash if main window was not found (caused crash on XP).
3. Fixed some logic bugs in the firmware update code.
4. Added code to calculate and check PEC codes for SMBus functions.
5. Fixed endian bug in u2aSMBUS_ReadWord() function.
6. Added handle checking code to u2aFEC_PurgeBuffer() and u2aFEC_GetResult() to prevent crash if USB2ANY is not open.
7. Several places, changed value of invalid handle from '0' to 'INVALID_U2A_HANDLE'
8. Several places, changed method of validating handle from 'h == 0' to 'IsValidU2A_HANDLE(h)'
9. Added code to validate handle in GetApiPointer() function
10. Added code to validate handle in u2aStatus_GetErrorCode() function
11. Fixed bug in u2aStatus_IsUSB2ANYConnected
12. Fixed bug in u2aGetErrorList
13. Fixed bug in u2aStatus_GetErrorCode
14. Improved error checking in u2aSMBUS_Control
15. Fixed return value bug in u2aGPIO_ReadState
16. Added u2aUART_DisableReceiver function
17. Improved error checking in u2aADC_ConvertAndRead
18. Fixed return value bug in u2aADC_ConvertAndRead
19. Fixed return value bug in u2aMSP430_WordRead

20. Fixed incorrect return value in u2aMSP430_ByteRead
21. Improved error checking in u2aFEC_PurgeBuffer
22. Improved error checking in u2aFEC_GetResult
23. Fixed bugs and improved error checking in u2aStatus_EVMDetect
24. Added u2aGetDebugLogPath function
25. Added u2aDigital_Capture, u2aDigital_GetData, and u2aDigital_GetStatus functions
26. Changed logic in u2aEnableDeviceDetect()
27. Added u2aSetDeviceDetectVidPid()
28. Changed logic in u2aEnableDeviceDetectMsg()
29. Removed check for (s_MsgWnd == NULL) in u2aStatus_IsUSB2ANYConnected()
30. Added code in u2aOpen() to handle s_bSuppressFirmwareCheck flag
31. In u2aSMBUS_ReceiveByte(), don't check PEC if I2C read failed
32. In u2aSMBUS_ReceiveByte(), use CalcCRC8() instead of CalcPEC to check PEC
33. In u2aSMBUS_ReadWord(), use CalcCRC8() instead of CalcPEC to check PEC
34. In u2aSMBUS_ReadBlock(), fixed bugs in PEC calculation
35. Added code to call u2aInvokeBSL when Firmware Loader dialog opens.
36. Changed SetReceiveTimeout() to return ERR_PARAM_OUT_OF_RANGE if the *milliseconds* parameter is less than 2 or greater than 100
37. Added code to handle Digital Capture functions
38. Modified LED_SetState to send the BlinkCode parameter
39. Added commands for Digital Capture
40. FirmwareLoader: Fixed bug that caused crash on second programming attempt
41. FirmwareLoader: Updated BSL code to version 00.07.88.38
42. FirmwareLoader: Increased delay before InitializeBSL(), to prevent "USB device was unplugged" error
43. FirmwareLoader: Added code option to have OnDeviceChange() ignore VID/PID
44. FirmwareLoader: Added CHiddenMsgWindow::SetDeviceID()
45. The firmware loader will now display "Firmware version 1.x.x.x is deprecated..." message before loading firmware older than v2.x.x.x.

v2.6.1.20 (beta)

1. Added macros INVALID_U2A_HANDLE and IsValidU2A_HANDLE() to help validate handles
2. Added functions: CU2AClassUART_DisableReceiver(),Digital_Capture(),Digital_GetData, and Digital_GetStatus().
3. Added LEDS enum for LED states.

v2.6.1.0

1. Requires firmware version 2.6.1.0, or later.
2. Added u2aSuppressFirmwareCheck() function.
3. Changed splash screen code to not display splash if main window was not found (caused crash on XP).
4. Fixed some logic bugs in the firmware update code.

5. Added code to calculate and check PEC codes for SMBus functions.
6. Fixed endian bug in u2aSMBUS_ReadWord() function.
7. Added error code ERR_DATA_CRC_FAILED.

v2.6.0.5 (beta)

1. Support for firmware version 2.6.0.5.
2. Change ERR_NO_EVM text from "No EVM" to "EVM is required for external power".
3. Added RFFE commands:
   Cmd_RFFE_RegZeroWrite
   Cmd_RFFE_RegWrite
   Cmd_RFFE_ExtRegWrite
   Cmd_RFFE_ExtRegWriteLong
   Cmd_RFFE_RegRead
   Cmd_RFFE_ExtRegRead
   Cmd_RFFE_ExtRegReadLong
4. SPI: To reduce confusion, the SPI_ClockPhase enums have been changed:
   From:
   SPI_Change_On_First_Edge = 0
   SPI_Change_On_Following_Edge = 1
   To:
   SPI_Capture_On_Trailing_Edge = 0
   SPI_Capture_On_Leading_Edge = 1
   The original enums are still supported, but have been deprecated.
5. SPI: To reduce confusion, the SPI_LatchPolarity type name has been changed to the new SPI_ChipSelectPolarity type.
6. SPI_LatchPolarity has been defined as a synonym for SPI_ChipSelectPolarity.
7. SPI: To reduce confusion, the SPI_ChipSelectPolarity enums have been changed:
   From:
   SPI_Low_To_High
   SPI_High_To_Low
   To:
   SPI_CS_Active_Low
   SPI_CS_Active_High
   The original enums are still supported, but have been deprecated.
8. Added SMBUS_PEC_CRC8 as a synonym for SMBUS_PEC_ON.
9. Added the following new API functions:
   u2aEnablePacketLogging()
   u2aSuppressSplash()
   u2aLogComment()
   u2aDAC_SetVoltage()
   u2aFEC_GetResult()
   u2aFEC_PurgeBuffer()
   u2aPower_ReadStatus()

10. Deprecated the following API functions:

GetSerialNumberFromHandle()	(replaced by u2a GetSerialNumberFromHandle)
u2aReadResponse()	(obsolete)

11. Fixed a bug in I2C_MultiRegisterWrite() that caused addressing problems.
12. Added a new Firmware Loader dialog that shows all available devices that can be updated and allows the user to select which one to update.
13. Added an "Update to firmware version" list box to the Firmware Loader dialog, which lists the available versions of firmware that can be loaded. The recommended version is automatically pre-selected, but the user can change the selection to an alternate version.
14. Fixed a bug that prevented multiple boards from being opened (and used) from a single instance of the USB2ANY.DLL library. The DLL now supports up to six simultaneous connections.
15. By default, USB2ANY.DLL displays a splash screen when a program attaches to it. The splash screen can be suppressed by calling the new u2aSuppressSplash()function with a TRUE parameter before calling other API functions.
16. Fixed a bug in the u2aFirmwareVersion_Read() function that erroneously caused it to return ERR_OPERATION_FAILED if the bufsize parameter was set less than 5. It now correctly returns ERR_BUFFER_TOO_SMALL if the bufsize parameter is set less than 4.

NOTE: The following changes are internal and not directly visible to end users.

17. Improvements to ByteFifo class.
18. Added line and function parameters to the ERROR_EXIT macro.
19. Changed GetLocalAppDataPath(), WriteSetting(), and ReadSetting() functions to use CSIDL_LOCAL_APPDATA instead of CSIDL_COMMON_APPDATA for data storage.
20. Added GetSDKPath() function;
21. Added MakeFirmwarePath() function.
22. Changed FindNewestFirmware() function to create a list of all available firmware versions, in addition to finding the newest version.
23. Added GetAvailableFirmwareVersions() functions to get the list of all available firmware versions.
24. Added a RunProcess() function to execute a child process.
25. Completely changed (and simplified) the code used to display the splash screen. The old code had several problems, including lock-ups and crashes.
26. USB2ANY_SDK.H: Fixed comments that showed the wrong ID numbers for several commands.
27. Added Cmd_InvokeBSL and Cmd_FirmwareDebugMode commands.
28. Added a BOOL bEnumOnly parameter to the u2aFindControllersBSL() function.
29. Changed communications code to use a CByteFifo for ADC results.
30. The enabler for packet logging is now persistent (previous logging state is remembered).
31. Command names are now shown in the packet log, instead of hex codes.
32. Received packets are now synchronized using semaphore objects.
33. Asynchronous "Payload" packet types are parsed in a CriticalSection.
34. TI_USB2ANY_Init() function call was removed from the U2A_ApiClass constructor. This was causing open handles to be lost.
35. Individual semaphore objects are now used for each interrupt "channel", instead of just a single semaphore for all channels.

36. Added an ADC_Read() function to get data received asynchronously.
37. Added code to ADC_Acquire() and ADC_AcquireTriggered() to flush the FIFO before sending the command to the firmware.
38. Changed ADC_GetStatus() to get buffered status that was received asynchronously instead of requesting status from the firmware.
39. Added ADC interrupt handler.
40. Changed the EXIT_FUNCTION macro to return a value.
41. Added EXIT_FUNCTION_PTR macro.
42. Added internal FirmwareLoaderDialog() function.
43. Added BOOL bEnumOnly parameter to the u2aFindControllersBSL() to allow it to be used to enumerate controllers in BSL mode, without displaying the Firmware Loader dialog.
44. Added a call to TI_USB2ANY_Init() to the u2aOpen() function. This will delay the display of the splash screen until an attempt is made to actually open a connection to a board.
45. Added code to u2aOpen() that prevents checking of the firmware version if g_nFirmwareDebugMode is set to FW_DEBUG_NO_CHECK.
46. Added code to display the board's serial number in the "firmware version … is not supported" message box.
47. Added code to display the board's serial number in the "newer version of firmware is available" message box.
48. All SMBus commands are now emulated by the DLL using I2C commands:
    u2aSMBUS_SendByte()
    u2aSMBUS_WriteByte()
    u2aSMBUS_WriteWord()
    u2aSMBUS_WriteBlock()
    u2aSMBUS_ReceiveByte()
    u2aSMBUS_ReadByte()
    u2aSMBUS_ReadWord()
    u2aSMBUS_ReadBlock()
    u2aSMBUS_Control()
    u2aSMBUS_GetEchoBuffer()
49. Fixed a bug that caused the u2aI2C_MultiRegisterRead() function to return one extra byte.
50. Fixed a bug in u2aFEC_Configure() that could prevent it from returning ERR_PARAM_OUT_OF_RANGE when an invalid parameters was passed to it.

KNOWN ISSUES:

51. The following functions are not implemented:
    u2aMSP430_MemoryRead()
    u2aMSP430_MemoryWrite()

v2.5.1

1. Support for firmware version 2.5.1.0.

v2.5.0

1. USB2ANY.dll now displays a splash screen on startup. BETA versions are labeled as such.
2. Improved error handling.
3. More detailed debug logging.
4. All API functions now return error codes returned by the firmware.
5. u2aFindControllers can now locate USB2ANY boards that are executing BSL code.
6. When u2aOpen() is called, it now checks the current firmware version of the attached controller and compares it with the newest known firmware version. If the firmware can be updated, the user is prompted to do so. If the current firmware version is less than the minimum required version, the update is mandatory (otherwise, the open will fail).
7. Fixed a bug in I2C_MultiRegisterRead() that caused it to return invalid data.
8. Fixed a bug in I2C_ReadInternal () that caused it to return invalid data and/or byte count.
9. Added additional baud rates to u2aUART_Control() function: 300, 320, 600, 1200, 2400, and 4800.
10. Added u2aADC_AcquireTriggered() function.
11. Removed all references to FEC1, FEC2, and FEC3, since the board has only one FEC port (FEC0).
12. Fixed some bugs in the FEC code and verified that it is working.
13. Added high-resolution profiling code to all API functions.
14. Added u2aEnableAPIProfiling() function to enable/disable API profiling.
15. Added u2aProfileTimestamp() function to allow a high-resolution timestamp to be inserted into the debug log at any time.
16. Added u2aPower_Notify() function to enable/disable an asynchronous callback function that will be called whenever a power event occurs.
17. Fixed bugs in u2aEasyScale_Read() function that caused incorrect data to be returned.
18. Changed name of u2aDAC_SetVoltage() API function to u2aDAC_SetValue(), to better match its purpose.
19. Changed name of u2aPower_SetVoltage() API function to u2aPower_SetVoltageRef(), to better match its purpose.

NOTE: The following changes (in blue) are internal and not directly visible to end users.

20. Added optional debug logging to HID driver.
21. Changed maximum size limit for packet logs to 50 Kbytes.
22. Logging to SysDebug now defaults to disabled.
23. Packet logging now defaults to enabled.
24. The single packet receive queue has been replaced by packet queues for each distinct command.
25. FEC responses are no longer queued, but are stored in a global structure.
26. Received data from u2aUART_Read operations is now stored in a 1024-byte FIFO buffer.
27. Packet log files are now prevented from growing uncontrolled. On initialization, the packet log is archived, and a new one started, if the file size is greater than 50K bytes.
28. New packet log sessions are now timestamped.
29. The format of packet log entries has been changed and now includes the sequence number, packet type, command, packet length, PEC, status code, payload length, and payload data (if applicable).
30. A callback handler has been added for power events.

31. Improved logging of packet read errors.
32. Added code to detect and report orphaned packets.
33. Added code to detect and report payload size errors.
34. Added handler for payload packets.
35. Cleaned-up SendBuffer function.
36. Improved error checking in SendCommandPacket function.
37. Added FEC response handler.
38. FindDevices function can now optionally ignore serial numbers. This is user to detect BSL firmware.
39. Many commands have been update to use the "new" protocol for communication between USB2ANY.DLL and the USB2ANY firmware.
40. Reliability improvements to ReadResponse function.
41. DACs_Write command now allows access to DAC2 (only for OneDemo).
42. DACs_Write now differentiates the DAC type (DAC5574 for OneDemo, DAC5571 for USB2ANY).
43. Added ADC_AcquireTriggered function for delayed triggering of the ADC.
44. Implemented EasyScale_Control, EasyScale_Write, and EasyScale_Read commands.
45. Implemented Power_Notify command to allow asynchronous power change notifications (via callback).

v2.1.0

1. API now supports plugging/unplugging of a USB2ANY.
2. Improvements to **u2aEnableDeviceDetect** API function.
3. Added **u2aIsUSB2ANYConnected** function to allow connection status to be obtained.
4. Fixed a bug that caused the **u2aPower_ReadStatus** function to fail.
5. Fixed a bug that caused the **u2aPower_ReadStatus** function to fail when in Polled Mode.
6. Fixed a bug that caused the **u2aI2C_RegisterRead** function to return wrong results.
7. Fixed a bug that caused the **u2aI2C_MultiRegisterRead** function to return wrong results.
8. Fixed a bug that caused the **u2aI2C_RawRead** function to return wrong results.
9. Modified **u2aGPIO_ReadPort** API function to return correctly when in Polled Mode.
10. Modified **u2aSPI_WriteAndRead_P** API function to return correctly when in Polled Mode.
11. Modified **u2aUART_Read** API function to return correctly when in Polled Mode.
12. Modified **u2aPower_ReadStatus** API function to return correctly when in Polled Mode.
13. Many improvements to error detection and logging.
14. Added **u2aEnableDebugLogging** API function.
15. Improvements to reliability and functionality of **u2aGetErrorList** API function.
16. Added **u2aStatus_GetErrorCode** API function, which returns a code for the first error that occurred during execution of the last called API function.
17. Added **u2aStatus_GetText** and **u2aStatus_GetTextW** (Unicode version) functions to get a string containing the text for an error code.
18. The **u2aGetStatusText** API function is now deprecated. Use **u2aStatus_GetText** instead.
19. A bug was fixed that prevent a debug log from being created if the **USB2ANY\Logs** sub-folder did not exist in the default Windows Documents folder. Now, the required sub-folder(s) will be created, if they do not already exist.

20. Added **u2aGetSerialNumberW** function to get a Unicode string containing the serial number of the specified board.
21. Added **u2aOpenW** function to open a controller using a Unicode string containing the serial number.
22. Various improvements to the **u2aReadResponse** API function.
23. By default, a new debug log file will be created when the SDK DLL is loaded. If a previous log file exists, with the same name, it will be saved with **.bak** appended to the name, overwriting any previous "**.bak**" file.
24. If append mode is enabled for debug log files, the default maximum file size is 1Mb. If the file grows beyond that limit, it will be saved and a new log file started.
25. A bug was fixed that could cause several extraneous log files (one for each running application that attached the DLL) to be opened when the SDK DLL was loaded in a debugger.

NOTE: The following changes (in blue) are internal and not directly visible to end users.

26. Fixed a bug that caused a crash if hid_enumerate() returned info for a device without a serial number.
27. Fixed a bug caused by a conflict with a definition of **IsLoggingEnabled** in the Windows include file UrlMon.h.
28. Changed several U2A_ApiClass member function names for consistency with the SDK function names and the documentation:

Old Name	New Name
I2C_Register_Write	I2C_RegisterWrite
I2C_Multi_Register_Read	I2C_MultiRegisterRead
I2C_Read_Internal	I2C_ReadInternal
I2C_Write_Internal	I2C_WriteInternal
I2C_Raw_Write	I2C_RawWrite
I2C_Raw_Read	I2C_RawRead
GPIO_Write_Control	GPIO_WriteControl
GPIO_Write_State	GPIO_WriteState
GPIO_Read_State	GPIO_ReadState
LED_Write_Control	LED_WriteControl

29. Fixed a bug that caused the **U2A_ApiClass:: Power_ReadStatus** function to fail.
30. Fixed a bug that caused the **U2A_ApiClass::I2C_RegisterWrite** function to fail because the first parameter was 8-bits, instead of 16-bits.
31. Increased size of internal buffer for the hexstring() function from 128 bytes to 512 bytes.
32. Added internal GetApiHandleFromSerialNumber() function to allow a U2A_HANDLE to be obtained from the serial number of an opened board.
33. Added **u2aSetResponseMode** API function. Only **RESP_SYNC** and **RESP_POLLED** modes are supported at this time. This function is currently supported only for use by **USB2ANY_API.DLL** and is not documented.

Known Issues:

34. Disabled **MultiControllerMode** for this release. This version supports only one USB2ANY controller per instance of USB2ANY.DLL.
35. The functions **GPIO_SetPort**, **GPIO_WritePort**, and **GPIO_ReadPort** are not implemented in this release. They will be implemented in a future release.
36. I2C transactions at 10kHz may have problems (e.g., missing STOP) when attempting to transfer more than 12 bytes of data. It is recommended that only very small packets (i.e., less than 10 bytes) are used when the bit rate is 10kHz. This problem will be addressed in a future release.

v2.0.8

1. The name of the FEC Interval enumerated constants has been changed from **FEC_Int** to **FEC_Interval** to prevent conflicts with other constants.
2. Added enumerated constants for **EasyScale_WriteSpeed**, **EasyScale_ReadSpeed**, **EasyScale_WriteACK**, **EasyScale_Status_Received**, **EasyScale_ReceivedDataStatus**, and **EasyScale_Threshold**.
3. Changed the name of a constant from **Command_EasyScale_Set** to **Cmd_EasyScale_Control**.
4. Changed the name of a constant from **Command_EasyScale_Write** to **Cmd_EasyScale_Write**.
5. Added constant **Cmd_EasyScale_Read**.
6. Fixed bug that could prevent **u2aFirmwareVersion_Read** from returning correct value.
7. **u2aGetStatusText** will now use an internal buffer if second parameter is NULL.
8. **u2aReadResponse** will now write received bytes to debug log.
9. **u2aI2C_RegisterWrite** was erroneously exported as **u2aI2C_Register_Write**.
10. **u2aI2C_RegisterRead** was erroneously exported as **u2aI2C_Register_Read**.
11. Fixed bug in **u2aI2C_RegisterRead** that prevented it from returning the correct value.
12. **u2aI2C_MultiRegisterWrite** was erroneously exported as **u2aI2C_Multi_Register_Write**.
13. **u2aI2C_MultiRegisterRead** was erroneously exported as **u2aI2C_Multi_Register_Read**.
14. Fixed bug in **u2aI2C_MultiRegisterRead** that prevented it from returning the correct value.

NOTE: The following changes (in blue) are internal and not directly visible to end users.

15. Added class functions **FEC_Control**, **FEC_CountAndRead**, **Interrupt_Control**, **EasyScale_Control**, **EasyScale_Write**, and **EasyScale_Read**.
16. Modifications for FEC functions.
17. Added code for USB plug/unplug detection (WIP).
18. Fixed bug in **u2aI2C_InternalRead** that could cause wrong return value in POLLED_MODE.
19. Fixed bug in **u2aI2C_RawRead** that could cause wrong return value in POLLED_MODE.
20. Fixed bug in **u2aGPIO_ReadState** that could cause wrong return value in POLLED_MODE.
21. Fixed bug in **u2aADC_ConvertAndRead** that could cause wrong return value in POLLED_MODE.
22. Fixed bug in **u2aMSP430_WordRead** that could cause wrong return value in POLLED_MODE.
23. Fixed bug in **u2aMSP430_ByteRead** that could cause wrong return value in POLLED_MODE.